

STUDY PLAN

PhD in Computer Science (AI/ML Specialization)

National Taiwan University of Science and Technology (NTUST)

Fall 2026 Admission

1. INTRODUCTION

My name is **Mulata Kebede**, an Ethiopian lecturer and researcher with over a decade of experience in computer science education and industry application. I hold a Master of Science in Computer Science from Addis Ababa University (GPA: 3.52) and a Bachelor of Science in Computer Science from Mekelle University (GPA: 3.47). I am applying for the PhD program in Computer Science at NTUST, specializing in **Artificial Intelligence, Deep Learning, and Natural Language Processing (NLP)**.

My long-term research ambition is to develop innovative AI solutions that address real-world challenges, particularly for low-resource languages and educational applications in developing contexts. NTUST's strong industry connections, particularly the ASUS Industry-Sponsored Scholarship, and its reputation in applied AI research make it the ideal environment to achieve these goals.

2. ACADEMIC BACKGROUND & RESEARCH EXPERIENCE

Master's Thesis Research

My master's thesis, "**Automatic Part of Speech Tagger for Ge'ez Language using Hybrid Approach**," represents my core research experience. Ge'ez is an ancient liturgical language with very limited digital resources, making it a typical example of a low-resource language in natural language processing. In this research, I:

- Investigated hybrid approaches combining rule-based methods with statistical machine learning techniques
- Addressed challenges of limited annotated corpora and complex morphological structures
- Developed algorithms to automatically tag parts of speech, contributing to digital preservation and linguistic analysis of this historic language

This work gave me hands-on experience with:

- NLP techniques for low-resource languages
- Machine learning model development and evaluation
- Linguistic feature engineering
- Research methodology and academic writing

Teaching Excellence

As a **Lecturer at Raya University** (2018-present) and previously at Dilla University, I have taught core computer science courses including:

- Advanced Database Management Systems
- Advanced Programming
- Data Structures and Algorithms
- Object-Oriented Programming
- System Programming

This teaching experience strengthened my understanding of core computer science concepts and strengthened my ability to communicate complex ideas—essential skills for doctoral research and future academia.

Industry Experience

My role as **Senior ICT Officer at Ethiopian Electric Utility** and **Application/Web Developer at NEAEA** provided practical experience in:

- Developing enterprise applications (ERP systems, Fleet Management System)
- Building web platforms serving thousands of users
- Stock Management System
- Grade 10 & 12 Result and Placement App (serving students nationwide)

These projects demonstrate my ability to **apply technical knowledge to solve real problems at scale**—a mindset I will bring to my PhD research.

Technical Skills

I have developed technical skills in:

- **Programming:** Python, Java, JavaScript, Kotlin
- **Web/App Development:** HTML, CSS, Flutter, Android
- **Databases:** SQL, NoSQL
- **Tools:** Git, GitHub, various cloud platforms
- **Certifications:** Big Data Engineer (IBM), Certified Ethical Hacker, Security and Compliance Fundamentals

3. WHY NTUST?

I have chosen NTUST for several compelling reasons:

1. Strong AI/NLP Research

NTUST's Department of Computer Science and Information Engineering has a growing reputation in artificial intelligence and natural language processing. The university's research output in these areas, combined with its focus on applied technology, aligns perfectly with my goal of developing practical AI solutions.

2. Industry Connections (ASUS Scholarship)

I am particularly interested in the **ASUS Industry-Sponsored Scholarship**. The opportunity to:

- Receive full tuition and monthly stipend
- Complete internships at ASUS, a global technology leader
- Gain priority employment consideration after graduation

This industry-academia bridge is invaluable. ASUS's work in AI-powered devices and intelligent systems provides a valuable opportunity to connect academic research with practical industrial applications.

3. Taiwan's Technology Ecosystem

Taiwan is a global hub for semiconductor and technology innovation. Studying at NTUST would immerse me in this ecosystem, exposing me to cutting-edge research, industry practices, and collaboration opportunities unavailable elsewhere.

4. Potential Faculty Advisors

After reviewing NTUST's faculty profiles, I am particularly interested in professors working on:

- **Natural Language Processing for low-resource languages** (directly aligned with my master's thesis)
- **Deep Learning applications in educational technology** (building on my teaching experience)
- **AI for social good and developing regions** (my long-term passion)

I would be honored to work with any faculty member whose research intersects with these areas.

4. PROPOSED RESEARCH PLAN

Research Area: AI for Low-Resource Languages and Educational Applications

My research experience with Ge'ez language processing, combined with my extensive teaching background, naturally leads me toward this interdisciplinary focus.

Research Questions

1. How can deep learning techniques be adapted for low-resource languages with complex morphology, like Ge'ez and other Ethiopian languages?
2. Can transfer learning and multilingual models effectively bridge the resource gap for under-represented languages?
3. How can AI-powered educational tools be designed to improve learning outcomes in developing countries with limited infrastructure?

Proposed Methodology

- **Phase 1 (Year 1):** Coursework, literature review, and identifying specific research gaps. Build connections with potential advisors and ASUS mentors.
- **Phase 2 (Year 2-3):** Develop novel neural architectures for low-resource NLP tasks. Explore techniques such as:
 - Transfer learning and cross-lingual model adaptation
 - Data augmentation for low-resource scenarios
 - Morphologically-aware neural networks
 - Few-shot and zero-shot learning approaches
- **Phase 3 (Year 3-4):** Apply research to practical domains:
 - Educational technology (intelligent tutoring systems, automated assessment)
 - Digital preservation of endangered languages
 - Collaboration with ASUS on AI product development
- **Phase 4 (Year 4):** Thesis completion, publication in top conferences/journals, and graduation.

Expected Contributions

1. Novel algorithms and models for low-resource language processing
2. Open-source datasets and tools for Ethiopian and other under-represented languages

3. AI-powered educational applications tailored for developing regions
4. Peer-reviewed publications at top venues (ACL, EMNLP, AAAI, etc.)

5. FUTURE CAREER GOALS

Short-term (During PhD)

- Master advanced deep learning and NLP techniques
- Publish research in top-tier international conferences and journals
- Complete meaningful internships at ASUS, contributing to real AI products
- Build a strong professional network in Taiwan's tech industry

Medium-term (post-PhD, 0-5 years)

- Work as a research scientist at a leading technology company (ideally ASUS in Taiwan)
- Continue research on AI for low-resource languages and educational applications
- Bridge the gap between cutting-edge AI research and practical deployment

Long-term (5-10+ years)

- Return to Ethiopia to establish a research center focused on AI for African languages and educational technology
- Train the next generation of Ethiopian AI researchers
- Collaborate with international partners (including NTUST and ASUS) to bring global expertise to local challenges
- Contribute to Ethiopia's digital transformation through AI innovation

6. CONCLUSION

My journey from teaching computer science fundamentals to researching NLP for Ge'ez language, from developing national examination platforms to aspiring for a PhD at NTUST, reflects a consistent commitment: **using technology to solve meaningful problems**. The PhD program at NTUST, with its unique industry integration through the ASUS Scholarship, represents the ideal next step in this journey.

I am confident that my research experience, technical skills, teaching background, and passion for AI-driven social impact make me a strong candidate. I am equally confident that NTUST's world-class faculty, resources, and industry connections will provide the environment I need to grow into a researcher who can bridge the gap between advanced AI and real-world needs in developing regions.

Thank you for considering my application. I look forward to the opportunity to contribute to NTUST's research community and to represent the university as a future leader in AI for social good.

Mulata Kebede

Date: 11/03/2026

Signature: 